

B. TECH
(SEM VI) THEORY EXAMINATION 2022-23
AUTOMOTIVE ELECTRICAL AND ELECTRONICS

Time: 3 Hours**Total Marks: 100****Note:** Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt all questions in brief.**2 x 10 = 20**

- (a) What are the benefits of wiring harness?
- (b) State Kirchhoff's law.
- (c) Briefly explain the working of rectifier.
- (d) State two advantages and two disadvantages of a permanent magnet starter.
- (e) Write short note on ballast resistance.
- (f) Draw a circuit diagram of a oil pressure warning light.
- (g) Write any four benefits of electrically driven air conditioning system.
- (h) What do you understand by micromechanics?
- (i) Briefly discuss about FlexRay.
- (j) What is an optoelectronic sensor?

SECTION B

2. Attempt any three of the following:**10x3=30**

- (a) Compare positive earthing to negative earthing system? Which one is preferred and why? Explain in detail.
- (b) Describe the working of balancing coil type of fuel gauge with neat sketch.
- (c) Discuss in detail about the various methods for adjustment of headlights.
- (d) Explain about bus system? Also mention its advantages & requirements.
- (e) Discuss about the following:
 - (i) Pressure Sensors
 - (ii) Position Sensors

SECTION C

3. Attempt any one part of the following:**10x1=10**

- (a) What do you understand by battery ratings? Explain any two types of commonly used battery ratings.
- (b) Write a note on following:
 - (i) Principle of operation of lead -acid battery
 - (ii) Sulphation failure

4. Attempt any *one* part of the following: 10x1=10

- (a) What is an Alternator? Compare it with Generator. Which one is preferred? Explain the reasons in detail.
- (b) Write the name of various types of starters? Explain any two in detail.

5. Attempt any *one* part of the following: 10x1=10

- (a) Explain the working of ignition system with ignition coil of 4-cylinder SI engine.
- (b) Discuss the important characteristics of ideal spark plug.

6. Attempt any *one* part of the following: 10x1=10

- (a) Explain ABS control unit in detail.
- (b) Describe the following: (i) ECM (ii) Steering control unit

7. Attempt any *one* part of the following: 10x1=10

- (a) Discuss in detail about acceleration & vibration sensors.
- (b) Write the note on the following:
 - (i) Stepper Motor
 - (ii) Immobilizer System